

Institute of Actuaries of India

Subject SA6 – Investment

April 2016 Examinations

INDICATIVE SOLUTIONS

Solution 1:**i) Attribution analysis of an active fund**

The active management effect is the sum of the selection, allocation, and interaction effects. You can use the active management effect to determine the amount the investment manager has added to a portfolio's return. If the active management effect is positive, the investment manager has contributed positively to the portfolio's return. If the active management effect is negative, the investment manager has not contributed positively to the portfolio's return.

The return decomposition model uses the following calculation to calculate the active management effect:

Allocation + Selection + Interaction = Active Management Effect

- The allocation effect measures an investment manager's ability to effectively allocate their portfolio's assets to various segments. A segment refers to assets or securities that are grouped within a certain classification such as Equity, Fixed, or Technology. The allocation effect determines whether the overweighting or underweighting of segments relative to a benchmark contributes positively or negatively to the overall portfolio return. Positive allocation occurs when the portfolio is over-weighted in a segment that outperforms the benchmark and underweighted in a segment that underperforms the benchmark. Negative allocation occurs when the portfolio is over-weighted in a segment that underperforms the benchmark and underweighted in a segment that outperforms the benchmark.

`[(portfolio weight) - (benchmark weight)]`

`[(benchmark return) - (benchmark total return)]`

- The selection effect measures the investment manager's ability to select securities within a given segment relative to a benchmark. The over or underperformance of the portfolio is weighted by the benchmark weight, therefore, selection is not affected by the manager's allocation to the segment. The weight of the segment in the portfolio determines the size of the effect—the larger the segment, the larger the effect is, positive or negative.

`[(benchmark weight)]`

`[(portfolio return) - (benchmark return)]`

- The interaction effect measures the combined impact of an investment manager's selection and allocation decisions within a segment. For example, if an investment manager had superior selection and over-weighted that particular segment, the interaction effect is positive. If an investment manager had superior selection, but underweighted that segment, the interaction effect is negative. In this case, the investment manager did not take advantage of the superior selection by allocating more assets to that segment.

`[(portfolio weight) - (benchmark weight)]`

`[(portfolio return) - (benchmark return)]`

[4]

ii) Equity attribution differs from debt attribution

- Equity Attribution - the excess return (positive or negative) is explained through Sector allocation and Security selection strategies.
- Debt Attribution - the excess return (positive or negative) is explained through the Duration and Credit (spread effect) strategies taken by the fund managers.

[2]

iii) Steps in asset allocation process for a pension fund

The overall process of asset allocation can be broken down into the following stages:

- a) Trigger – a decision is made to review the current asset allocation, perhaps following a particular event like legislation, economic conditions, change in liability profile, etc.
- b) Objective setting - The starting point for the asset allocation process is to set suitable financial objectives for the pension scheme. The critical point is that financial objectives must be determined before the number crunching and analysis commences. The financial objectives drive the asset allocation and not the other way round.
- c) Analysis – An asset liability model is a tool to model likely future developments of the pension scheme, and is therefore a powerful aid in investigating the implications of different asset allocations.
- d) Decision – the results of the analysis are presented to the trustees or an investment sub-committee and a decision on the preferred alternative is made.
- e) Detailed decision on the chosen alternative - outperformance targets, the mix between active and passive management, review of fund manager, the use of tactical asset allocation, etc.
- f) Manager agreements will need to be drafted, agreed and signed. The same will need to be done for the statement of investment principles to reflect the new asset allocation, which will need to be circulated to the investment managers.
- g) Implementation of the asset allocation strategy.
- h) Performance measurement relative to the strategic benchmark.
- i) Monitoring the on-going appropriateness of the strategic benchmark itself.

[4]

iv) The reasons for pension schemes undertaking an AL study are:

- Understanding and quantifying investment risk within the scheme like:
 - o Security – What are the causes, likelihood and possible size of any deficit?
 - o Contributions – What level and variability of contributions might the company and members expect?
 - o Benefits – If discretionary benefits are a possibility then what level and variability might be expected?
 - o Accounting measures – What P&L and balance sheet variability can be expected and how is this affected by different accounting standards?
 - o Apportionment – What is the degree to which risk is apportioned between different stakeholders (for example, pension scheme beneficiaries, stockholders, advisors, regulators, etc.)?

- Once the trustees and company have understood the degree of risk they are running they will often want to use an AL study to assist in amending or setting a strategic asset allocation by comparing alternate assets.
- It may not always be possible to achieve an appropriate outcome through amending the strategic asset allocation alone. A study may therefore go on to assist in amending or setting scheme benefit design. The study would help in determining what impact scheme design can have on security, contributions, benefits and accounting measures. For instance, if the company moves from DB to a DB and DC hybrid, in order to share risk with the employees, what range of benefits might be expected?
- Clearly the pace of funding will have an impact on security and there can also be an interaction between the funding policy and the investment strategy. Therefore, the trustees and company may also wish to understand the consequences of alternative funding strategies.
- One could use AL studies to understand the consequences of changes in the economic or investment environment that may alter the risk profile of the fund
- AL studies may be used to assist in setting up a new scheme. The new group of trustees and employer will want to take advice on how to invest the transferred assets and/or future contributions.
- Once the investment strategy has been put in place AL studies may be of further use to assist in setting appropriate manager structure arrangements.

[4]

v) A broad tracking error range vs. a specific target

Benefits

- The active risk is typically swamped by the strategic risk. The overall solvency of the fund is not particularly sensitive to changes in the active risk level. For example, if the investment manager were to have an extreme level of underperformance in any one year then the impact of this is likely to be swamped by the absolute performance of the strategic asset allocation.
- This allows the investment manager some freedom to increase or reduce the risk level based on the conviction of their views. This increases the scope for the manager to outperform.
- The prospective tracking error of the fund is not directly observable or controllable. Any estimate of the future tracking error is subject to a high degree of error. Focusing in on a specific point target is therefore spurious and doomed to failure.

Concerns:

- Although pension funds invest for the long-term it is human nature to look at any short-term performance information available and be overly concerned by periods of short-term underperformance.
- Investment managers are concerned with managing their own business risk. If they feel that trustees will react badly to periods of short-term underperformance then their self-preservation instincts will lead to index hugging.
- A narrow tracking error range can encourage the manager to adopt a myopic focus and an over-zealous approach to risk management. In this scenario risk considerations can drive portfolio decisions and force turnover in order to maintain the target risk level. This will not only incur transaction costs but may also distract the manager from their key goal of adding value.

[3]

vi) Control process to monitor whether the fund is being managed in line with the risk profile:

- Buy-in from senior management. It is critical that senior management have an understanding of portfolio risk and have approved a formal portfolio risk policy.
- A specialist portfolio risk management department. This team will be central to the risk management process.
- A clear process for identifying and reporting on whether a fund is being managed in line with its risk profile.
- A clear statement of the actions that need to be taken when a fund is outwith its risk range and the responsibilities for ensuring that these actions are followed through.
- On-line controls to stop inappropriate deals going through. For example this could be based on stock/sector/country limits plus client restrictions (e.g. no tobacco).
- A regular reporting process (e.g. monthly) to ensure that funds are within appropriate risk ranges taking into account all relevant measures.
- Portfolio risk management should be integrated in the investment process such that fund managers have an in-depth understanding of portfolio risk and are aware of the risk that they are taking when they are placing deals.

[3]

vii) Limitations of a liability driven benchmark

Liability driven benchmark is based on expected liability cashflows. The differences between the actual and expected cashflows mean that strictly we cannot perfectly match the liabilities. This could be due to:

- Mortality risk - This includes fitting given population to one or more model(s) or standard table(s) and anticipation of future trends.
- Salary increases - In theory, the sponsoring employer should be able to control this element of risk. But in practice, decisions over salary increases typically do not take into account the consequent effect on pension liabilities.
- Cash commutation - This is a material risk, if commutation factors are not directly related to the underlying investments, as members have the option to crystallise a substantial portion of their entitlements.
- Transfers-out (including bulk transfers)
- Withdrawals from service
- Retirement timing (including ill health retirements)
- Death before retirement
- Taxation/regulation
- Expenses including future inflation
- Deferred revaluation
- Higher costs involved in 'perfect' matching

[4]

viii)

Compare of different liability driven benchmark options:

	Market index	Immunised bond portfolio	Cashflow matching	Cashflow matching
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			using bonds	using bonds and swaps
Instruments	Bonds within index	Longest dated government backed bonds	Bond of all terms and the possibility for some strips	Bonds and swaps or a swap portfolio backed by cash
Interest Rate Risk	Benchmarking against market indices results in very poor matching for most schemes. A typical scheme may have duration of 20+ years but a long bond index duration is about 11.	Typically about 80% of interest rate risk can be removed through duration management if sufficiently long dated stock is available. Immunisation leaves a trustee exposed to non-parallel moves in the yield curve - e.g. changes in shape	Can vary - holding shorter bonds to match early year cashflows can shorten duration so some extra long bullet bonds may be needed to ensure the first moment is matched.	Interest rate risks fully hedged out to longest dated available (or liquid) swap
Inflation Risk	Though we can invest in a mix of fixed interest and index-linked instruments, the duration is likely to be short.	Though we can invest in a mix of fixed interest and index-linked instruments, the maximum duration is likely to be 20-25 years.	Cash flows can be split between inflation sensitive cash flows and cash flows fixed in nominal terms and matched separately.	Cash flows can be split between inflation sensitive cash flows and cash flows fixed in nominal terms and matched separately. LPI swaps could be used to match capped and floored increases
Mortality Risk	No cover	No cover	No cover	No cover
Credit Risk	No credit risk in a gilt	In theory could use credit but	Larger schemes could	Interest rate swaps may

	portfolio but credit can be introduced by using a corporate index	there is limited diversification available in very long-dated sterling corporate bonds.	construct a reasonably diversified credit portfolio, particularly in respect of shorter duration liabilities, but smaller schemes would suffer undue issuer concentration.	introduce counterparty risk but this risk can be almost fully mitigated through daily marking to market and appropriate controls on collateral
Liquidity Risk	Very low	Very low	Can vary - getting to a high level of matching through physicals rather than derivative instruments could be achieved through less liquid strips.	Swaps are not particularly liquid beyond 50 years and there may be uncertainty in regards to future liquidity for reversing out swap contracts.
Operational Risk	Very low, can use pooled funds	Low, although a segregated account will be required	Segregated account required and some problems assessing manager performance	Difficult to ascertain at less liquid long end. Maintenance of collateral account required
Costs	Low	Segregated account may make this approach uneconomical for small schemes but not a costly solution for larger	Initial costs may be high if strips are not economically priced. Segregated account may make this approach uneconomical	Swap costs will depend on terms from investment banks that can vary a lot. Segregated account may make this approach

		schemes.	for small schemes but not a costly solution for larger schemes.	uneconomical for small schemes but not a costly solution for larger schemes.
Complexity/ Monitoring	Simple	Simple but bespoke monitoring	Monitoring of asset cash flow profile against that of liabilities.	Monitoring of asset cash flow profile against that of liabilities. Trustee education required to understand swap use

[16]

ix) Legal and regulatory requirements and challenges for use of swaps in India.

Legal requirements

- Ensure that the pension scheme and trustees are authorised to use swaps, or not prohibited from doing so. This may require modification to the existing Trust Deed and the Statement of Investment Principles.
- The counterparties need to agree on the legal documentation for the trade and the collateral arrangements.
- Documentation - ISDA and two way CSA
- Corporate Governance - While dealing with such potentially complex products the Board and the senior management of insurer should understand the nature of the risk undertaken, complexities involved, stress levels etc.

Regulatory requirements

- Allowed Instruments - Forward Rate Agreements, Interest Rate Swaps, Exchange traded Interest Rate Futures
- Allowed Purpose - Derivatives must only be used for hedging purposes - reinvestment of maturity proceeds, investment of interest income receivable, expected premium income receivable.
- Allowed Counterparties - Commercial Banks and Primary Dealers permitted by RBI.
- Exposure Limits - as per IRDA Investment Regulations. No contract to be entered with promoter group entities.
- Size cap - Outstanding notional principal amount should not exceed 100% of the book value of the fixed income investments.

Challenges in India

- IRS markets in India are not developed because of the lack of participation of the Insurers who can be big buyers for IRS. If allowed, banks can hedge their floating rate liabilities and insurers their fixed interest liabilities by entering into IRS.

- Limited counterparties
- Collateral management
- Hedging vs. Speculation: Difficult to ensure that IRS are only being used as hedges and not speculative
- Notional amount exposed to IRS needs to be curtailed to a fixed percentage of underlying assets
- Heavy accounting / disclosure requirements

[4]

x) Counterparty risk and Collateralization

- One major concern for pension schemes entering into swap contracts is the exposure that they take to the banking counterparty. Equally, the bank will need to consider its credit exposure to the fund.
- Normally, swap contracts are written so that the value of the contract is zero at outset - i.e. the market value of the fixed / inflation linked flows is equal to that of the floating / nominal flows. However, as yields and inflation expectations change over time the swap will have a value to the pension scheme, or to the bank.
- The resulting credit exposure is normally managed by collateral process. The counterparty that is out-of-the-money provides security in the form of collateral to the party who is in-the-money.
- Collateral is normally in the form of either cash or high-quality securities (e.g. government bonds). Securities with a lower credit rating will be subject to a haircut when assessing the amount that needs to be set aside.
- The collateral will typically be held in a separate earmarked account with the custodian.
- Need to agree on a number of defined parameters regarding the operation of the collateral, such as the Threshold, Minimum Transfer Amount, Rounding Amount and the Valuation Dates. These parameters control the frequency at which collateral is exchanged and the amount of collateral that is posted.
- At each Valuation Date the mark-to-market value of the swap is calculated and agreed. Any mark-to-market exposure below the Threshold is not collateralised. Any exposure above the Threshold not covered by existing collateral requires a transfer of collateral. However, for practical reasons, collateral is only transferred if the amount required exceeds the Minimum Transfer Amount. Finally, the amount of collateral transfer is rounded according to the Rounding Amount.
- A small residual exposure remains, in particular for any exposure between the last Valuation Date and default, and for any threshold or minimum transfer amounts. There is a trade-off between minimising credit risk and practical convenience for both parties.
- The pension scheme will need to ensure that its custodian is able to deal with the administration required.

[4]

xi) Advantages and disadvantages of insuring the liabilities

Advantages

- The (vast majority of) risk of meeting the benefits is passed on to the insurance company.

- Trustees no longer have to be concerned with whether the sponsor will continue to be able to meet any shortfall of assets against liabilities.
- Members have a high degree of certainty that their benefits will be paid.
- Sponsor is no longer exposed to volatility in funding (contributions), or the corporate balance sheet.

Disadvantages

- It is unlikely that an insurance policy would precisely match a scheme's benefit arrangement where the value of the benefit to be paid may be influenced by the sponsor, trustees, or members.
- Insurance companies are subject to statutory reserving requirements. These requirements impact on both the capacity of insurance companies to take on new business and the terms for accepting business the premium has to cover the cost of capital as well as the expected value of the benefits payable.
- The insurance companies active in the buyout market are proprietary companies that conduct business with the aim of making a profit.
- The mismatch between the trustees investment policy with that of the insurer leads to fluctuations between the insurance premium and the assets held and high reinvestment costs.

[4]

xii) When it is likely to be preferable to buyout liabilities:

For schemes that are either closed to new joiners or have ceased providing future service benefit accrual and broken the link between accrued benefits and future salaries, it is preferable to buyout liabilities. At some point the cost of the scheme's upkeep (investment, administration, accounting, valuation, documentation etc.) will mean that it is no longer economically viable to keep the scheme going. The trustees may opt to buy out the liabilities rather than run the scheme on. This avoids potential accusations that would follow if the scheme's experience were such that benefits had to be cut back by more than on immediate buyout.

[2]

xiii)

How credit rating agency rates insurance companies.

- CRISIL's uses Financial Strength Rating (FSR) as an indicator of the insurance company's financial capacity to meet policyholders' obligations.
- There are two broad categories - Secure from "AAA" to "BBB", Vulnerable from "BB" to "D".
- FSR does not indicate timeliness of payment or the likelihood of the use of a defence such as fraud to deny claims.
- FSR does not a guarantee of financial strength or a recommendation to purchase or discontinue a certain policy.
- It does not refer to an insurance company's ability to meet non-policy obligations such as obligations on debt contracts.
- CRISIL's methodology includes an assessment of the following aspects:
 - o Inherent industry risk relative to other industries
 - o Analysis of the business and financial performance
 - o Effect of parent/group support on the standalone ratings of the insurance company, if applicable

- o Identification of Rating Sensitivity Factors (RSF) and crystallization of Outlook
- The key factors affecting the claims paying ability are:
 - o Industry Risk: While arriving at a company's business risk rating, CRISIL factors in the risks associated with the external environment, that is the insurance industry in which the company operates.
 - o Business Risk: Company is assessed on parameters that drive its relative business position in the insurance industry.
 - o Financial Risk: Company is assessed on parameters like capitalization and profitability that determine its relative financial strength vis-à-vis other players in the industry.
 - o Parentage: This parameter evaluates the parental support that is available to an entity.

[4]

xiv)

Counterparty credit risk for life insurance companies vs. general insurance companies

Credit risk is less of an issue for most Life companies since (a) they are less reliant upon reinsurance and (b) what reinsurance they have tends to be with very strong reinsurers. As such they do not suffer quite the same problems as some general insurance companies, hence the more relaxed approach. Many approaches are taken to account for credit risk in the context of reserve setting.

- Do nothing as the default probabilities of the (very strong) reinsurers is so low that it lies beyond the threshold of the VAR test used for capital, and as such has no impact on reserve calculations.
- Use a higher discount rate to reflect the risks - apply bigger discounts to more distant recoveries.
- Use a basic 'percentage of notional' write-down approach to make an entirely subjective allowance.
- Ignore the smaller exposures and focus only on the larger sums at risk.
- Treat the reinsurance cashflows as a defaultable corporate bond and follow the statutory capital rules for those.
- Do 'what if' scenario testing, such as market shocks and/or failure of the biggest reinsurer, and use this to determine a provision.
- Collateral arrangements

[4]

xv) India is offering various necessary conditions for hedge funds like

- Second market liquidity
- Increased use of derivatives - futures, options
- Capital raising in US and UK home markets has grown tougher since the global financial crisis.
- Comparatively lower management fees
- Investment can be made through FII thus attracting more HNIs to enter India through these
- Doesn't charge anything on performance or profit on the fund thus giving India a plus point from others.

[3]

xvi)

Advantages and disadvantages of a hedge fund:

Advantages -

- Diversification when used along with traditional investments (negatively or un-correlated)
- Allow short-selling thus help diversify a 'long-only' portfolio
- Reduced risk/ enhanced returns

Disadvantages -

- Lack of proper past data including survivorship bias and back-fill bias. Data must be adjusted with appropriate caveats before interpreting results
- Active return (alpha) therefore fund manager selection is very crucial
- Higher operational risk
- High fees
- Lack of proper regulation
- Lower liquidity and less transparency
- Non-normal returns (fat tailed and negatively skewed) - one bad year can wipe off a number of good ones

[3]

xvii)

Corporate pension plans find hedge funds to be an attractive investment:

- Better Asset/Liability Management: Hedge funds are more suited for asset/liability or "funding gap" driven approach.
- Private equity may compound at higher rates of return over long periods (given their substantial equity market risk compared to hedge funds), but they also introduce more risk into the equation as their higher volatility increases the potential mismatch with a plan's funded status.
- The outlook on gilt yields is volatile. There has been an increasing trend of pension plans utilizing hedge funds as an alternative to traditional fixed income allocations.
- Hedge funds can help in disciplined rebalancing of portfolios when the economic conditions change.
- The customizable and diverse nature of hedge funds allows one to adapt the overall risk exposures to the needs of a pension plan. From decreasing interest rate sensitivity to focusing on quickly deploying capital to tactical opportunities, hedge funds can provide additional investment levers.

[2]

[70 Marks]

Solution 2:

i) Alternative asset classes or specialist asset class are the non-conventional asset class that exclude asset classes like fixed interest securities, equities etc. The main examples of the asset classes are:

- private equity
- hedge fund investment
- credit derivatives
- infrastructure investment
- ETFs
- Structured products.

Other alternative asset classes include:

- securitisation
- Repos
- swaptions
- swaps
- Private debt.

[3]

ii)

Investment in other assets as specified in Regulations cannot exceed 15% of investment assets. Other assets include asset backed securities, alternate investment fund and/or direct investment into private equity (maximum 3%), unrated securities (maximum up to 5%). Investment in securitized assets (Mortgage based securities/ Asset based securities/ Securities receipts (SR)) both under approved and other investment category shall not exceed 10% of Investment Assets in case of Life companies. Approved investment in MBS/ABS with underlying Housing or Infrastructure asset shall not exceed 10% of investment assets in case of Life companies. Any MBS/ABS with underlying housing or infrastructure asset, if downgraded below AAA or equivalent shall be reclassified as Other investment. Investment in real estate can be done upto maximum of 5% in addition the 15% in alternate or other assets.

[2]

iii)

A life insurance company has long term liabilities, and most of the assets are held in fixed interest securities. The life insurance companies will follow asset liability matching strategies and only a low proportion of assets will be held under alternative asset class.

The insurance company may wish to hold investment in alternative asset class:

Potential for higher returns, possibly from increased beta,

Market inefficiencies,

Diversification due to a lack of correlation with existing assets or by exposure to underlying risks that are uncorrelated so reducing the overall portfolio risk.

[4]

iv)

It has been argued that commodity futures should be considered as “short term equities”. They are real assets whose value is determined by short term economic factors rather than expectations over the longer term. This gives some diversification from the traditional institutional real assets of property and equity shares.

The role of commodities as a diversifying asset class is that:

- commodities offer significant real returns that are produced by doing real economic work within the economy
- the returns accrue to the long-only investor without the need for active management, and
- commodities offer a package of diversification benefits unlike any other asset

Specifically, in those environments that have produced the worst results from financial assets — rising inflation, excessive global demand, supply disruptions — commodities have produced higher returns than any other asset class used by institutional investors. More importantly, those returns have been based on real underlying economics, suggesting that a similar pattern of returns is likely to recur in the future. Unlike other asset classes, commodities are concerned with short-term supply and demand and short-term risk.

Although the arguments presented above would seem to imply that commodity futures are a suitable investment for institutional investors they are not widely held by institutions.

Those who do not see commodities as attractive would argue that there is no strong historical evidence for a real return from commodities and that the markets are volatile, being driven by a number of factors unrelated to the underlying economic factors that affect institutional liabilities.

[6]

v) State ways in which one could invest in commodities.

Should an institution wish to gain exposure to commodity price movements it can do so in 3 ways:

- Invest in the underlying commodity (or basket of commodities)
- Commodity derivatives which are widely traded on major exchanges on either single commodities or an index.
- Invest in companies whose share price is influenced by commodity prices such as oil and mining companies.

[3]

vi) Outline the benefits and disadvantages of each method. (6)

Comment on management costs and skills, minimum bargain size, scope for diversification, basis risks with derivatives, volatility, liquidity and physical settlement/storage/shipping/transportation.

Holding individual contracts introduces risk of being delivered against

Disadvantages if companies are used as a proxy for commodity investment:

- It is unlikely that there will be exposure to just one commodity.
- The company's management may alter the exposure via acquisitions or disposals or by hedging its position.
- The company's share price may be influenced by other factors.

- The company will incur various operating expenses which will dilute the overall return.
- Use of commodity shares (mining, exploration companies) gives less diversification from equity market than physical would.

[6]

vii) Assuming no arbitrage argument, how the value of commodity futures can be derived similar to one can used for gilt and equity index futures.

In the valuation of commodity futures a distinction has to be drawn between commodities which are commonly held as investment assets and those which are held for commercial purposes.

For commodities, such as precious metals, which are held as investments, no-arbitrage arguments can be used to show that the value of a future must be given by a formula very similar to the one used for gilt and equity index futures,

Future price = spot price of underlying commodity + cost of carry.

Here the cost of carry is the financing cost of holding the underlying commodity, plus storage costs.

[2]

viii) Please explain the terms contango market , convenience yield and backwardation.

The cost of carry is the financing cost of holding the underlying commodity, plus Storage costs. Because the cost of carry is positive, the futures price is normally above the spot price. This is known as a contango market.

In the case where the underlying commodity is not widely held as an investment there is no such simple mathematical relationship between the futures price and the spot price. The usual situation is that there is a positive value to ownership of the physical commodity (e.g.as a protection against future shortages or in order to be able to take advantage of them by selling at a high price). This value is described as the convenience yield of the commodity and the formula becomes

Future price = spot price + cost of carry convenience yield.

When the convenience yield is higher than the cost of carry the futures price will be below the spot price, a situation known as backwardation.

[4]

[30 Marks]
